Thermodynamics and Thermochemistry

Thermodynamics: The study of the flow of heat or any other form of energy into or out of a system as it undergoes a physical or chemical transformation is called Thermodynamics.

Scope of Thermodynamics:

(1) Most of the important laws of Physical Chemistry, including the van‟t Hoff law of lowering of vapour pressure, Phase Rule and the Distribution Law, can be derived from the laws of thermodynamics.

(2) It tells whether a particular physical or chemical change can occur under a given set of conditions of temperature, pressure and concentration.

(3) It also helps in predicting how far a physical or chemical change can proceed, until the equilibrium conditions are established.

SYSTEM, BOUNDARY, SURROUNDINGS

A system is that part of the universe which is under thermodynamic study and universe is surroundings.

The real or imaginary surface separating the system from the surrounding boundary.

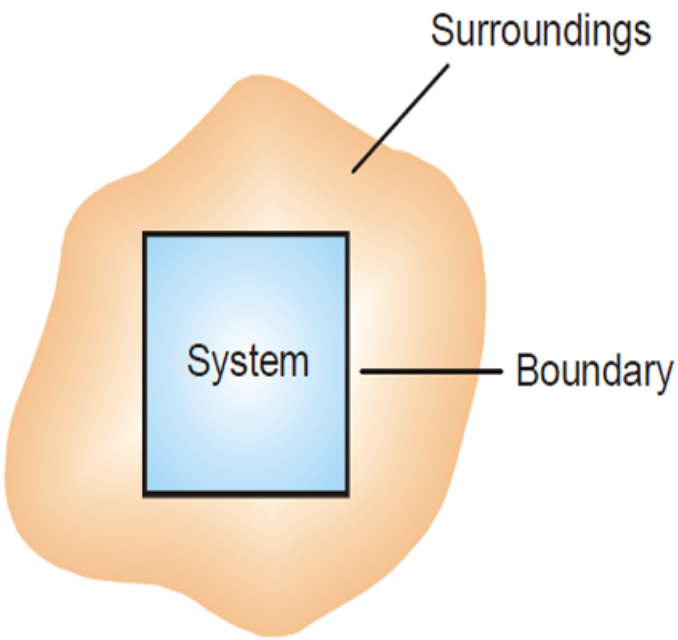


Figure 7.1
Thermodynamic system.

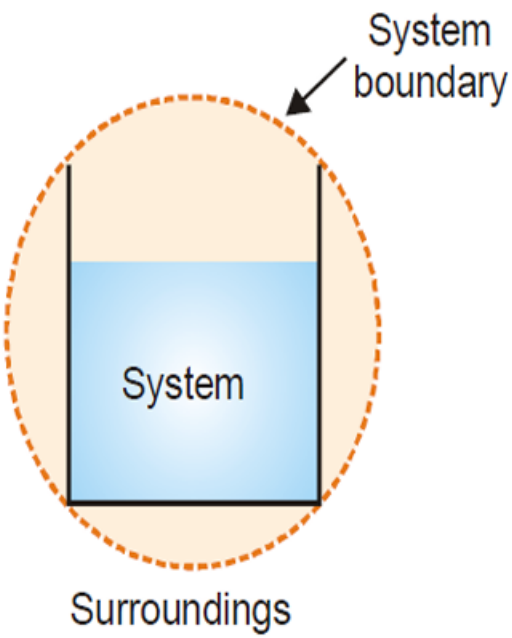


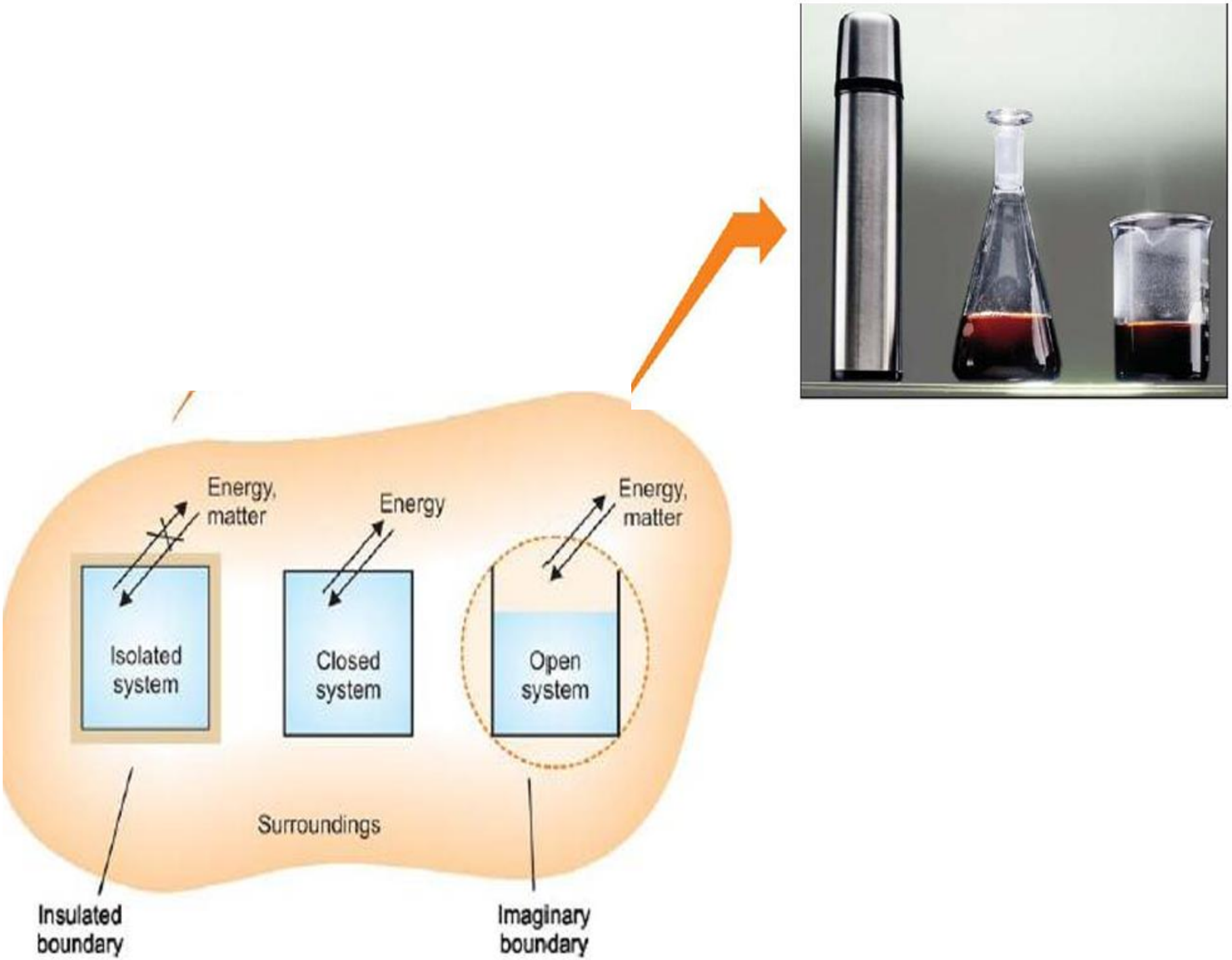
Figure 7.2
Water contained in a beaker constitutes a system.

Types Of Thermodynamic Systems:

(1) Isolated System: When the boundary is both sealed and insulated, no interaction is possible with the surroundings. Therefore, an isolated system is one that can transfer neither matter nor energy to and from its surroundings.

(2) Closed System: Here the boundary is sealed but not insulated. Therefore, a closed system is one which cannot transfer matter but can transfer energy in the form of heat, work and radiation to and from its surroundings.

(3) Open System: In such a system the boundary is open and un-insulated. Therefore, an open system is one which can transfer both energy and matter to and from its surroundings.


Energy, matter
Energy
Energy, matter
Isolated system
Closed system
Open system
Surroundings
Insulated boundary
Imaginary boundary

Thermodynamic Processes: When a thermodynamic system changes from one state to another, the operation is called a Process. These processes involve the change of conditions (temperature, pressure and volume).

The various types of thermodynamic processes are :

(1) Isothermal Processes: Those processes in which the temperature remains fixed, are termed isothermal processes. This is often achieved by placing the system in a thermostat (a constant temperature bath).For an isothermal process $dT= 0$

(2) Adiabatic Processes: Those processes in which no heat can flow into or out of the system, are called adiabatic processes. Adiabatic conditions can be approached by carrying the process in an insulated container such as „thermos" bottle. High vacuum and highly polished surfaces help to achieve thermal insulation. For an adiabatic process $dq= 0$

(3). Isobaric Processes: Those processes which take place at constant pressure are called isobaric processes. For example, heating of water to its boiling point and its vaporisation take place at the same atmospheric pressure. These changes are, therefore, designated as isobaric processes and are said to take place isobarically. For an isobaric process $dp = 0$

(4). Isochoric Processes: Those processes in which the volume remains constant are known as isochoric processes. The heating of a substance in a non-expanding chamber is an example of isochoric process. For isochoric processes $dV = 0$.

(5). Cyclic Process: When a system in a given state goes through a number of different processes and finally returns to its initial state, the overall process is called a cycle or cyclic process. For a cyclic process $dE = 0$, $dH = 0$.

Reversible And Irreversible Processes: A thermodynamic reverse process is one that takes place infinitesimally slowly and its direction at any point can be reversed by an infinitesimal change in the state of the system.

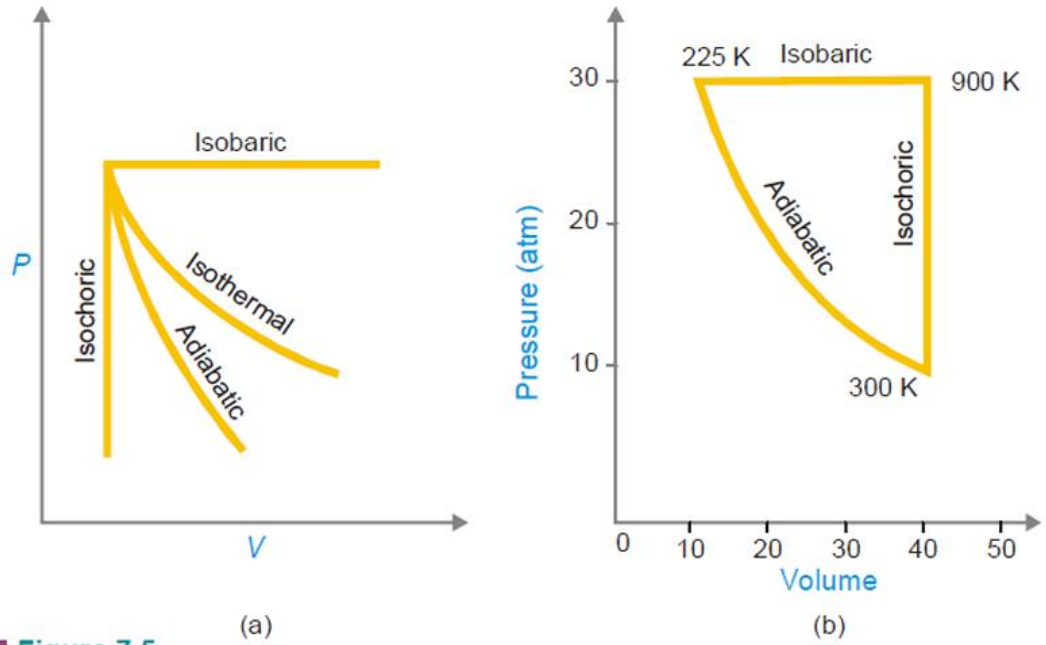


Figure 7.5
(*a*) Four basic thermodynamic processes; (*b*) A cyclic process.

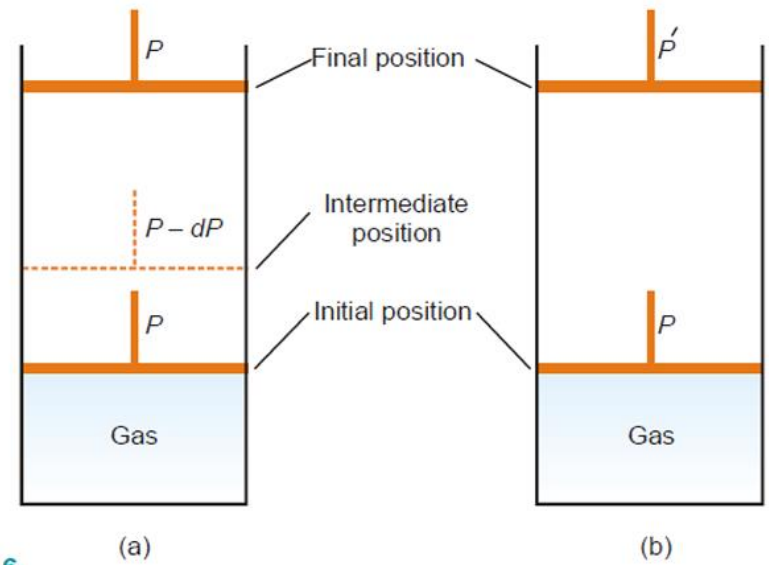


Figure 7.6
(*a*) Reversible expansion occurs by decreasing the pressure on the piston by infinitesimal amounts. (*b*) Irreversible expansion occurs by sudden decrease of pressure from P to P′, when the gas expands rapidly in a single operation.

When a process goes from the initial to the final state in a single step and cannot be carried in the reverse order, it is said to be an irreversible process.

Here the system is in equilibrium state in the beginning and at the end, but not at points in between. Consider a certain quantity of a gas contained in a cylinder having a weightless and frictionless piston. The expansion of the gas can be carried by two methods illustrated in Fig. 7.6. Let the pressure applied to the piston be P and this is equal to the internal pressure of the gas. Since the external and internal pressures are exactly counterbalanced, the piston remains stationary and there is no change in volume of the gas. Now suppose the pressure on the piston is decreased by an infinitesimal amount dP. Thus the external pressure on the piston being P – dP, the piston moves up and the gas will expand by an infinitesimal small amount. The gas will, therefore, be expanded infinitely slowly i.e., by a thermodynamically reversible process. At all stages in the expansion of the gas, dP being negligibly small the gas is maintained in a state of equilibrium throughout. If at any point of the process the pressure is increased by dP, the gas would contract reversibly. On the other hand, the expansion is irreversible (Fig. 7.6 b) if the pressure on the piston is decreased suddenly. It moves upward rapidly in a single operation. The gas is in equilibrium state in the initial and final stages only. The expansion of the gas, in this case, takes place in an irreversible manner.

DIFFERENCES BETWEEN REVERSIBLE AND IRREVERSIBLE PROCESSES

Reversible Process	Irreversible Process
1. It takes place in infinite number of infinitesimally small steps and it would take *infinite time* to occur.	1. It takes place *infinite time*.
2. It is *imaginary* as it assumes the presence of frictionless and weightless piston.	2. It is *real* and can be performed actually.
3. It is in equilibrium state at *all stages* of the operation.	3. It is in equilibrium state only at the *initial and final stages* of the operation.
4. All changes are *reversed* when the process is carried out in reversible direction.	4. After this type of process has occurred all changes *do not return* to the initial state by themselves.
5. It is extremely slow.	5. It proceeds at *measureable speed*.
6. Work done by a reversible process is *greater* than the corresponding irreversible process.	6. Work done by a irreversible process is *smaller* than the corresponding reversible process.

Internal Energy: The total of all the possible kinds of energy of a system, is called its Internal Energy. The word „internal" is often omitted and the energy of a system always implies internal energy. The internal energy of a system, like temperature, pressure, volume, etc., is determined by the state of a system and is independent of the path by which it is obtained. **Hence internal energy of a system is a state function**. For example, we consider the heating of one mole of liquid water from 0º to 100º C. The change in energy is 1.8 kcal and is the same regardless of the form in which this energy is transferred to the water by heating, by performing work, by electrical means or in any other way. The internal energy of a system is represented by the symbol E. It is neither possible nor necessary to calculate the absolute value of internal energy of a system. In thermodynamics we are concerned only with the energy changes when a system changes from one state to another. If ΔE be the difference of energy of the initial state (Ein) and the final state (Ef), we can write $\Delta E = Ef - Ein$

ΔE is +ve if Ef is greater than Ein and –ve if Ef is less than Ein. A system may transfer energy to or from the surroundings as heat or as work, or both.

Laws of Thermodynamics

The Three laws of Thermodynamics

The first laws of thermodynamics, also called conservation of energy. We can use this knowledge to determine the amount of energy in a system, the amount lost as waste heat and the efficiency of the system.

First Law of Thermodynamics: Also known as the law of energy conservation, it states that energy cannot be created or destroyed in an isolated system. The first law is often expressed in the form of the equation

$$\Delta U = Q - W$$

where ΔU is the change in internal energy of the system, Q is the heat added to the system, and W is the work done by the system on its surroundings. This law essentially quantifies the principle of conservation of energy.

The second law of thermodynamics states that the disorder in the universe always increases. As the disorder in the universe increases, the energy is transformed into less usable forms. Thus, the efficiency of any process will always be less than 100%.

The Second Law of Thermodynamics : This law introduces the concept of entropy, a measure of disorder or randomness in a system. The second law states that in an isolated system, the total entropy can never decrease over time.

The third law of thermodynamics tell us that all molecular movement stops at a temperature we call absolute zero, or 0 Kelvin (-273 deg C). Since temperature is a measure of molecular movement, there can be no temperature lower than absolute zero. At this temperature, a perfect crystal has no disorder.

The Third Law of Thermodynamics : A system's entropy approaches a constant value as its temperature approaches absolute zero.

More typically in engineering applications we focus on the 1st law of thermodynamic

First Law Of Thermodynamics: The first law of thermodynamics is, in fact, an application of the broad principle known as the Law of Conservation of Energy to the thermodynamic system. It states that: the total energy of an isolated system remains constant though it may change from one form to another.

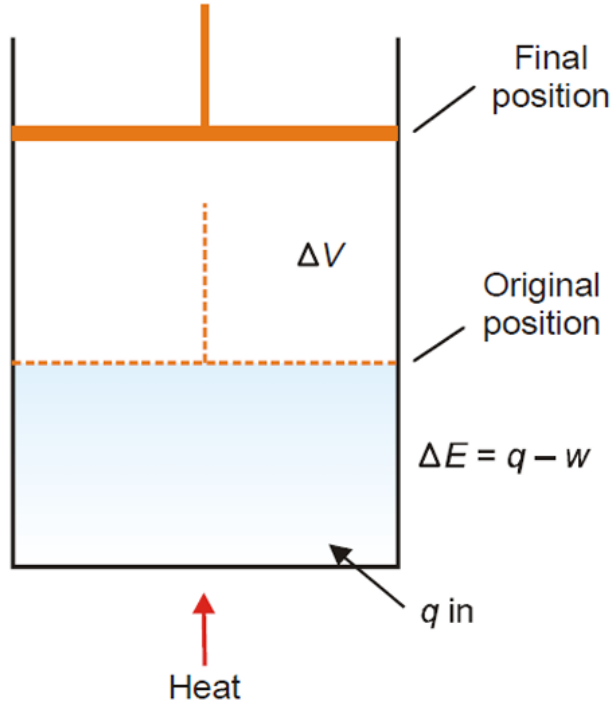


Figure 7.11
Illustration of First Law. Heat adds to internal energy, while work subtracts.

When a system is changed from state A to state B, it undergoes a change in the internal energy from EA to EB.

Thus, we can write $\Delta E = EB - EA$ This energy change is brought about by the evolution or absorption of heat and/or by work being done by the system. Because the total energy of the system must remain constant,

we can write the mathematical statement of the First Law as : $\Delta E = q - w$ ...(1) where q = the amount of heat supplied to the system, w = work done by the system Thus First Law may also be stated as :the net energy change of a closed system is equal to the heat transferred to the system minus the work done by the system. To illustrate the mathematical statement of the First Law, let us consider the system „expanding Hot gas" (Fig. 7.11).

The gas expands against an applied constant pressure by volume ΔV. The total mechanical work done is given by the relation- $w = P \times \Delta V$ ...(2) From (1) and (2), we can restate $\Delta E = q - P \times \Delta V$

SOLVED PROBLEM 2. 1g of water at 373 K is converted into steam at the same temperature. The volume of water becomes 1671 ml on boiling. Calculate the change in the internal energy of the system if the heat of vaporisation is 540 cal/g.

SOLUTION

As the vaporisation takes place against a constant pressure of 1 atmosphere, work done for an irreversible process, w, is

$$w = P(V_2 - V_1)$$
$$= nRT$$
$$= \frac{1}{18} \times 1.987 \times 373$$
$$= 41 \text{ cal/g}$$

R=1.987 cal/mol/K

Now $q = 540 \text{ cal/g}$

Since $\Delta E = q - w$ (First Law)

$$= 540 - 41$$

$\therefore$ $\Delta E = \mathbf{499\ cal/g}$

SOLVED PROBLEM 3. A gas contained in a cylinder fitted with a frictionless piston expands against a constant external pressure of 1 atm from a volume of 5 litres to a volume of 10 litres. In doing so it absorbs 400 J thermal energy from its surroundings. Determine ΔE for the process.

SOLUTION

$$\Delta E = q - w \quad \text{...(1)}$$

Here $q = 400 \text{ J}$

$$w = -P(V_2 - V_1) = -(1)(10 - 5)$$
$$= -5\ l \text{ atm}$$
$$= -506 \text{ J} \quad [\because 1\ l \text{ atm} = 101.2 \text{ J}]$$

Substituting values in (1)

$$\Delta E = 400 \text{ J} - (-506 \text{ J})$$
$$= 400 \text{ J} + 506 \text{ J}$$
$$= \mathbf{906\ J}$$

Enthalpy of a System: In a process carried at constant volume (say in a sealed tube), the heat content of a system is the same as internal energy (E), as no PV work is done. But in a constant-pressure process, the system(a gas) also expends energy in doing PV work. **Therefore, the total heat content of a system at constant pressure is equivalent to the internal energy E plus the PV energy.**

Hence Total heat content of a system is known as Enthalpy(Greek en = in; thalpos= heat) of the system and is represented by the symbol H.

Thus enthalpy is defined by the equation : $H = E + PV$

Free Energy (G): The available energy of a system used to perform work. It can be mathematically expressed as $G = H - TS$

Change of Free Energy can be expressed as $\Delta G = \Delta H - T\Delta S$

Heat Capacities:

If q calories is the heat absorbed by mass m and the temperature rises from T_1 to T_2, the heat capacity (c) is given by the expression

$$c = \frac{q}{m \times (T_2 - T_1)} \quad ...(1)$$

Thus heat capacity of a system is the heat absorbed by unit mass in raising the temperature by one degree (K or ° C) at a specified temperature.

When mass considered is 1 mole, the expression (1) can be written as

$$C = \frac{q}{T_2 - T_1} = \frac{q}{\Delta T} \quad ...(2)$$

where *C is denoted as Molar heat capacity.*

The molar heat capacity of a system is defined as the amount of heat required to raise the temperature of one mole of the substance (system) by 1 K.

Since the heat capacity (C) varies with temperature; its true value will be given as

$$C = \frac{dq}{dT}$$

Molar Heat Capacity at Constant Volume

According to the first law of thermodynamics

$$dq = dE + PdV \qquad ...(i)$$

Dividing both sides by dT, we have

$$\frac{dq}{dT} = \frac{dE + PdV}{dT} \qquad ...(ii)$$

At constant volume $dV = 0$, the equation reduces to

$$C_V = \left(\frac{dE}{dT}\right)_V$$

Thus the heat capacity at constant volume is defined as the rate of change of internal energy with temperature at constant volume.

Molar Heat Capacity at Constant Pressure

Equation (*ii*) above may be written as

$$C = \frac{dE}{dT} + \frac{PdV}{dT} \qquad ...(iii)$$

We know $\qquad H = E + PV$

Differentiating this equation w.r.t T at constant pressure, we get

$$\left(\frac{dH}{dT}\right)_p = \left(\frac{dE}{dT}\right)_p + P\left(\frac{dV}{dT}\right)_p \qquad ...(iv)$$

comparing it with equation (*iii*) we have

$$C_p = \left(\frac{dH}{dT}\right)_p$$

Thus heat capacity at constant pressure is defined as the rate of change of enthalpy with temperature at constant pressure.

Relation Between C_p and C_v

From the definitions, it is clear that two heat capacities are not equal and C_p is greater than C_v by a factor which is related to the work done. At a constant pressure part of heat absorbed by the system is used up in increasing the internal energy of the system and the other for doing work by the system. While at constant volume the whole of heat absorbed is utilised in increasing the temperature of the system as there is no work done by the system. Thus increase in temperature of the system would be lesser at constant pressure than at constant volume. Thus **C_p is greater than C_v.**

We know $$C_p = \frac{dH}{dT} \quad ...(i)$$

and $$C_v = \frac{dE}{dT} \quad ...(ii)$$

By definition $H = E + PV$ for 1 mole of an ideal gas

or $$H = E + RT \quad (\because PV = RT)$$

Differentiating w.r.t. temperature, T, we get

$$\frac{dH}{dT} = \frac{dE}{dT} + R$$

or $$C_p = C_v + R \quad \text{[By using equations } (i) \text{ and } (ii)\text{]}$$

or $$C_p - C_v = R$$

Thus C_p is greater than C_v by a gas constant whose value is 1.987 cal K^{-1} mol^{-1} or 8.314 J K^{-1} mol^{-1} in S.I. units.

SOLVED PROBLEM 2. Calculate the amount of heat necessary to raise 213.5 g of water from 25° to 100°C. Molar heat capacity of water is 18 cal mol^{-1} K^{-1}.

SOLUTION

By definition

$$C = \frac{q}{T_2 - T_1}$$

or

$$q = C(T_2 - T_1) \quad \text{– for 1 mole}$$

$$q = n\,C(T_2 - T_1) \quad \text{– for } n \text{ moles} \qquad ...(1)$$

In the present case

$$n = 213.5/18$$

$$C = 18 \text{ cal mol}^{-1} \text{ K}^{-1}$$

$$T_2 - T_1 = (373 - 298)\text{K}$$

Substituting the value in (1)

$$q = \frac{213.5}{18} \times 18 \times (373 - 298)$$

$$= \mathbf{16{,}012 \text{ cals} = 16.012 \text{ kcals}}$$

Joule-Thomson Effect: The phenomenon of producing lowering of temperature when a gas is made to expand adiabatically from a region of high pressure into a region of low pressure, is known as Joule-Thomson Effect or Joule-Kelvin Effect.

Joule-Thomson Experiment The apparatus used by Joule and Thomson to measure the temperature change on expansion of a given volume of gas is illustrated in Fig. 7.12. An insulated tube is fitted with a porous plug in the middle and two frictionless pistons A and B on the sides. Let a volume V1 of a gas at pressure P1 be forced through the porous plug by a slow movement of piston A. The gas in the right-hand chamber is allowed to expand to volume V2 and pressure P2 by moving the piston B outward. The change in temperature is found by taking readings on the two thermometers.

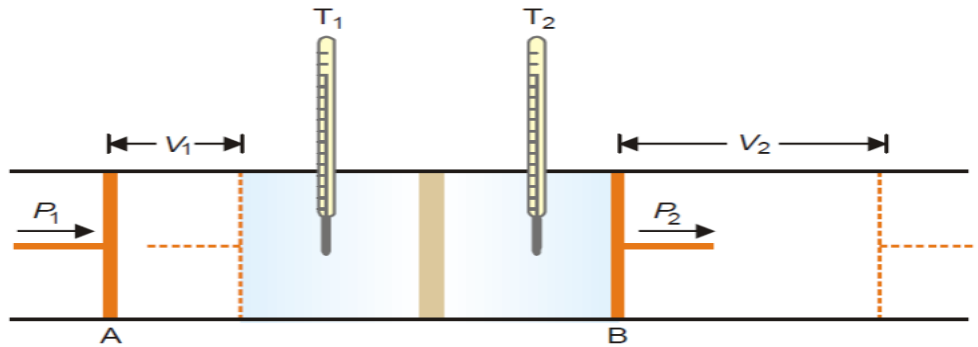


Figure 7.12
Joule-Thomson experiment.

Explanation: The work done on the gas at the piston A is $P1V1$ and the work done by the gas at the piston B is $P2V2$. Hence the net work(w) done by the gas is

$w = P2V2 - P1V1$

$\Delta E = q - w$ (First Law)

But the process is adiabatic and, therefore, $q = 0$

$\square \Delta E = E2 - E1 = -w = -(P2V2 - P1V1)$

or $E2 - E1 = -(P2V2 - P1V1)$

Rearranging,

$E2 + P2V2 = E1 + P1V1$

$H2 = H1$ or $\Delta H = 0$

Thus the process in Joule-Thomson experiment takes place at constant enthalpy.

Explanation of Joule-Thomson Effect:

We have shown above that Joule-Thomson expansion of a gas is carried at constant enthalpy. But

$H = E + PV$

Since H remains constant, any increase in PV during the process must be compensated by decrease of E, the internal energy. This leads to a fall in temperature *i.e.*, $T_2 < T_1$. For hydrogen and helium PV decreases with lowering of pressure, resulting in increase of E and $T2 > T1$. Below the inversion temperature, PV increases with lowering of pressure and cooling is produced.

Thermochemistry: Thermochemistry is the branch of physical chemistry which deals with the thermal or heat changes caused by chemical reactions. When the internal energy of reactants (Er) is greater than the internal energy of the products (Ep), the difference of internal energy, ΔE, is released as heat energy.

ΔE = Eproducts – Ereactants or ΔE = Ep– Er Such a reaction is called exothermic reaction. If the internal energy of the products (Ep) is greater than that of the reactants (Er), heat is absorbed from the surroundings. Such a reaction is called endothermic reaction. The amount of heat released or absorbed in a chemical reaction is termed the heat of reaction

Kirchoff'sequations : The heat of reaction changes with change in temperature of a gas due to variation in its specific heat. The equations representing the variation of heat change of reaction with temperature are known as Kirchoff's equations. At constant volume, the heat of reaction, ΔE, is given by the relation ΔE = E2 – E1 Where, E1 and E2 are the internal energies of the reactants and products. Differentiating this equation with respect to temperature at constant volume, we get

$$\left[\frac{d(\Delta E)}{dT}\right]_v = \left(\frac{dE_2}{dT}\right)_v - \left(\frac{dE_1}{dT}\right)_v \quad ...(1)$$

But we have already seen that

$$\left(\frac{dE}{dT}\right)_v = C_v$$

$$\therefore \quad \frac{d(\Delta E)}{dT} = (C_v)_2 - (C_v)_1 = \Delta C_v \quad ...(2)$$

where $(C_v)_2$ and $(C_v)_1$ are heat capacities of the products and reactants respectively. Or, **change in heat of reaction at constant volume per degree change in temperature is equal to the difference in heat capacities at constant volume of products and reactants.**

Integrating the above equation between temperatures T_1 and T_2, we have

$$\Delta E_2 - \Delta E_1 = \int_{T_1}^{T_2} \Delta C_v \, dT$$

or

$$\Delta E_2 - \Delta E_1 = \Delta C_v [T_2 - T_1] \quad ...(3)$$

where ΔE_2 and ΔE_1 are heats of reaction at temperatures T_2 and T_1 respectively.

Similarly, at **constant pressure** the heat of reaction ΔH is given by the reaction

$$\Delta H = H_2 - H_1$$

where H_2 is the heat content (enthalpy) of the products and H_1 being that of the reactants.

Differentiating with respect to temperature at constant pressure, we have

$$\left(\frac{d(\Delta H)}{dT}\right)_P = \left(\frac{dH_2}{dT}\right)_P - \left(\frac{dH_1}{dT}\right)_P \quad ...(4)$$

According to the equation, Chapter 7, we have

$$\left(\frac{dH}{dT}\right)_P = C_P$$

$$\therefore \quad \left(\frac{d(\Delta H)}{dT}\right)_P = (C_P)_2 - (C_P)_1 = \Delta C_P \quad ...(5)$$

where $(C_P)_2$ and $(C_P)_1$ are the heat capacities of products and reactants respectively.

or $$d(\Delta H) = \Delta C_P \times dT$$

Change in heat of reaction at constant pressure per degree change of temperature is equal to difference in heat capacities of products and reactants at constant pressure.

Integrating the equation between temperature T_1 and T_2, we have

$$\Delta H_2 - \Delta H_1 = \int_{T_1}^{T_2} \Delta C_p \, dT$$

or $$\Delta H_2 - \Delta H_1 = \Delta C_p [T_2 - T_1] \quad ...(6)$$

The relations (2), (3), (5) and (6) were first derived by Kirchoff and are called Kirchoff'sequations.These equations may be used for calculating heat of reaction at a given temperature when it is knownat some other temperature and when the heat capacities of products and reactants are known.

SOLVED PROBLEM 2. The heat of reaction $N_2 + 3H_2 \rightarrow 2NH_3$ at 27°C was found to be –21.976 kcal. What will be the heat of reaction at 50°C ?

The molar heat capacities at constant pressure and at 27°C for nitrogen, hydrogen and ammonia are 6.8, 6.77 and 8.86 cal mol^{-1} $degree^{-1}$.

SOLUTION

Here,

$$\Delta H = -21.976 \text{ kcal}$$

$$T_2 = 273 + 50 = 323 \text{ K}$$

$$T_1 = 273 + 27 = 300 \text{ K}$$

$$[T_2 - T_1] = (323 - 300) \text{ K} = 23 \text{ K}$$

ΔC_p = Heat capacities of products – Heat capacities of reactants

$$= (2 \times 8.86) - [6.8 + (3 \times 6.77)]$$

$$= 17.72 - (6.8 + 20.31) = -9.39 \text{ cal}$$

$$= -9.39 \times 10^{-3} \text{ kcal}$$

Substituting these values in Kirchoff's equation, we have

$$\Delta H_2 = \Delta H_1 + (T_2 - T_1)\, \Delta C_p$$

$$= -21.976 + [23 \times (-9.39 \times 10^{-3})]$$

$$= -21.976 + (-0.216) = -22.192 \text{ kcal}$$

$\therefore$ Heat of reaction at 50°C is = **22.192 kcal**

Hess's law: If a chemical change can be made to take place in two or more different ways whether in one step or two or more steps, the amount of total heat change is same no matter by which method the change is brought about.

The law also follows as a mere consequence of the first law of thermodynamics. Let us suppose that a substance A can be changed to Z directly.

$A \rightarrow Z + Q1 \qquad \Delta H1 = -Q1$

where $Q1$ is the heat evolved in the direct change. When the same change is brought about in stages :

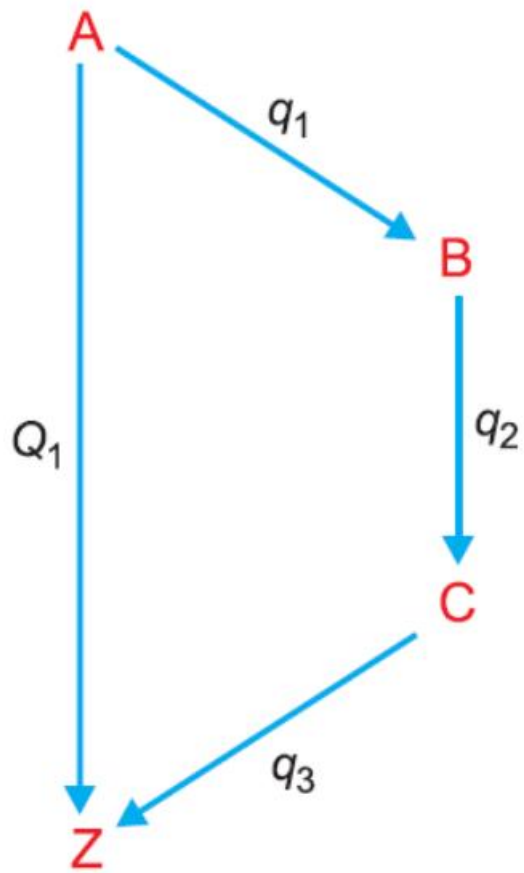
$A \rightarrow B + q_1 \qquad \Delta H2 = -q_1$

$B \rightarrow C + q_2 \qquad \Delta H2 = -q_2$

$C \rightarrow Z + q_3 \qquad \Delta H2 = -q_3$

the total evolution of heat $= q1 + q2 + q3 = Q2$

According to Hess's law $Q1 = Q2$. If it be not so, let $Q2 > Q1$. Then by transforming A to Z through stages and retransforming directly back to A, there would be gain of heat energy $= Q2 - Q1$. By repeating the process again and again an unlimited heat energy will be developed in an isolated system. This goes against the first law of thermodynamics. Hence $Q1$ must be equal to $Q2$.



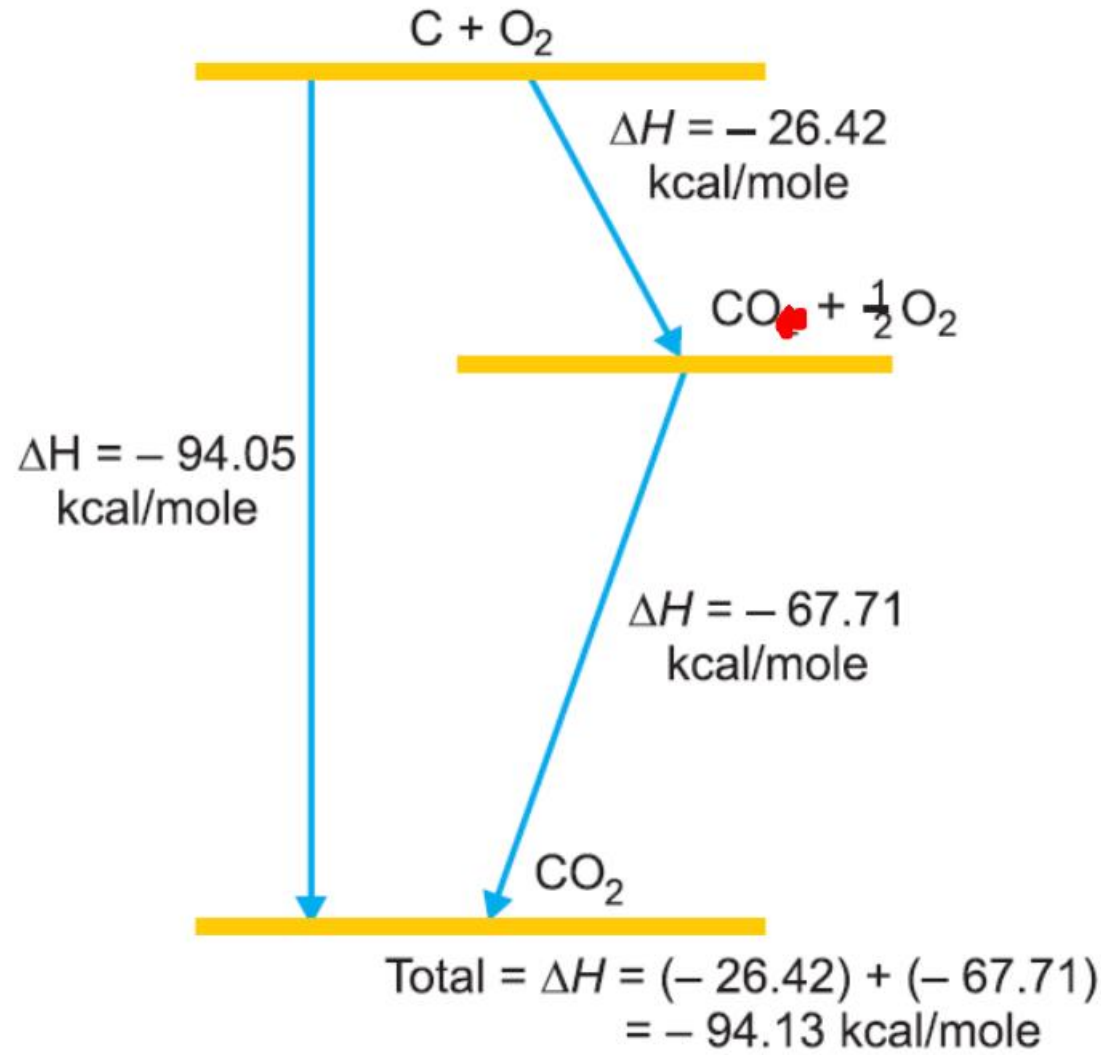


Figure 8.2
Illustration of Hess's Law.

Hess's law has been tested experimentally and shown to be true.

Illustrations of Hess's Law

(1) **Burning of carbon to CO_2.** Carbon can be burnt to carbon dioxide directly or it may first be changed to carbon monoxide which may then be oxidised to carbon dioxide.

1st way :

$$C(s) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -94.05 \text{ kcal}$$

2nd way :

$$C(s) + \tfrac{1}{2}O(g) \longrightarrow CO(g) \qquad \Delta H = -26.42 \text{ kcal}$$

$$CO(g) + \tfrac{1}{2}O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -67.71 \text{ kcal}$$

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overall change $\quad C(s) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -94.13 \text{ kcal}$

Bond Energy:The bond energy is defined as the average amount of energy required to break all bonds of a particular type in one mole of the substance.

Thus the bond energy of H – H bond is the energy required to break all the bonds in one mole of the gas. It is expressed in kcal mol^{-1} or kJ mol^{-1}. For example, the bond energy of H – H bond is 433 kJ mol^{-1} or 103.58 kcal mol^{-1}.